Multilevel Prototype Alignment for Cross-Domain Few-Shot Hyperspectral Image Classification

Hanchi Liu, Jinrong He, Yuhang Li, and Yingzhou Bi

***Abstract*— Hyperspectral image (HSI) classification holds significant application value in precision agriculture, environmental monitoring, and other fields. However, the high cost of large-scale labeling limits the widespread application of deep learning methods. Utilizing a large amount of labeled HSI data for auxiliary training of cross-domain few-shot learning (FSL) methods is an effective means to address this issue. Yet, in practical applications, spectral and spatial features vary across different scenarios, posing significant challenges to the generalizability and accuracy of cross-domain learning models. To tackle this problem, this article proposes a multilevel prototype alignment (MLPA) method for cross-domain few-shot classification, which adjusts feature representations by implementing multilevel feature alignment strategies at various hierarchical levels of the feature extraction network. This approach achieves fine-grained alignment of the source and target domain feature distributions, effectively reducing domain shift and enhancing the model's generalization capability on target domain data. Furthermore, by employing class prototype-based domain adversarial training, the method aligns the prototypes of the source and target domains without simply aligning the entire feature space, thus avoiding overlap in the feature distribution of different classes within the domain and mitigating negative transfer. The MLPA method effectively enhances the generalizability and discriminative power of features in the target domain, thereby improving the performance of cross-domain HSI classification. Experimental results demonstrate that MLPA outperforms other cross-domain few-shot HSI classification methods. Our source code is available at https://github.com/hejinrong/MLPA.**

***Index Terms*— Cross-domain few-shot learning (FSL), hyperspectral image (HSI) classification, meta-learning, prototype alignment.**

I. Introduction

HYPERSPECTRAL images (HSIs) are a special type of image data, which, compared to traditional color images, contain richer spectral information, typically including dozens to hundreds of continuous spectral bands. This information can be used to identify and classify different substances and materials [1]. As a result, their applications are very extensive, covering various fields such as remote sensing [2], [3], [4], food quality control [5], [6], [7], agriculture [8], [9], [10], environmental monitoring [11], [12], and military applications [13], [14], [15].

With the development of deep learning technology, significant achievements have been made in HSI classification based on deep learning, thanks to its powerful feature extraction capabilities and handling of high-dimensional data. Chen et al. [16] were the first to apply deep learning technology to HSI classification, proposing a hybrid framework combining principal component analysis (PCA), the deep learning architecture stacked autoencoder (SAE), and logistic regression. Due to the advantages of convolutional neural networks (CNNs) in capturing local image features and spatial hierarchical structures, subsequent research has focused on applying CNNs to HSI classification. Hu et al. [17] used 1D-CNN and leveraged spectral information for HSI classification, but 1D-CNN ignored spatial structural information. To fully capture the spectral and spatial features of HSIs, Kanthi et al. [18] proposed a 3D-CNN architecture. Zhong et al. [19] proposed a spectral–spatial residual network (SSRN), which sequentially learns discriminative features from the rich spectral information and spatial context in HSIs through spectral and spatial residual blocks. Lee and Kwon [20] proposed an innovative context deep CNN, which jointly extracts local spatial–spectral features through multiscale filter banks and achieves pixel-level classification of HSIs using a fully convolutional network. Although CNN-based HSI classification methods excel in feature representation, they have limited capabilities in obtaining deep semantic features and learning local sequential information of spectral features. Sun et al. [21] proposed the spectral–spatial feature tokenization transformer (SSFTT) method, which effectively captures low- and high-level semantic features of HSIs by combining spectral–spatial convolutional feature extraction and a transformer encoder module. Hong et al. [22] proposed the SpectralFormer method, which introduces the transformer architecture to learn local sequential information from adjacent bands and reduces the loss of information in deep network propagation through cross-layer skip connections. Although the aforementioned methods have made significant achievements in improving the accuracy and efficiency of HSI classification, they rely on a large number of labeled samples, and the cost of manually annotating a large number of hyperspectral samples is quite high.

To address the issue of insufficient labeled samples, cross-domain learning methods have been applied in HSI

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classification. These methods leverage a source domain dataset with abundant labeled samples to assist in training a target domain dataset that has only a few labeled samples, thereby enhancing the classification performance of the target dataset. However, because the source and target domain datasets are often two unrelated datasets with significant differences in data distribution, this can lead to domain shift issues [23]. Consequently, various cross-domain learning methods have emerged in the field of HSI classification, which can be broadly categorized into two types: fine-tuning-based methods and domain adaptation methods.

Fine-tuning-based methods involve training a model with strong representational capabilities on the source domain dataset and then fine-tuning the model on the target domain to adapt to new class data. Jiang et al. [24] pretrained a 3-D separable residual convolutional network on a source domain with ample samples, transferred the trained model parameters to the target domain, fine-tuned them with a small number of target domain samples, and then classified the remaining unlabeled target domain samples. The pretraining and fine-tuning strategy can reduce overfitting in CNNs, but when encountering inconsistent spectral characteristics and data scarcity in the hyperspectral domain, Lee et al. [25] proposed a universal cross-domain model to address this challenge, significantly improving accuracy and training efficiency, and successfully solving the overfitting problem, allowing the model depth to be increased to 13 layers. Although these fine-tuning methods effectively mitigate the challenges posed by domain differences, if the differences between the source and target domains are too great, the fine-tuning process may not fully achieve the effective transfer of pretrained knowledge to the target task, thereby limiting the enhancement of model performance [26].

Compared to fine-tuning methods, domain adaptation focuses on more fully exploring the similarities between the source and target domains, and more flexibly applying these similarities in the target domain tasks, enabling the knowledge acquired from the source domain to play a greater role in the target domain tasks. This approach narrows the gap between the source and target domains in the feature representation space [27]. Peng et al. [28] proposed a discriminative transfer joint matching (DTJM) method that successfully enhances the performance of cross-domain land cover classification for hyperspectral remote sensing images by minimizing the mean difference, instance reweighting, and preserving the local manifold structure of the features from the source and target domains in an unsupervised learning environment. Huang et al. [29] introduced a novel adversarial domain adaptation framework named calibrated prototypes and dynamic instance convolution (CPDIC), which effectively improves the performance of cross-domain HSI classification by combining static and dynamic feature extraction along with calibrated discriminator and prototype losses, surpassing existing unsupervised domain adaptation algorithms. The aforementioned methods effectively address the domain shift issue, but if the category labels of the source and target domains are inconsistent or not fully aligned, domain adaptation methods may struggle to correctly map and adapt the features, leading to negative transfer [30].

Addressing the issue of new categories in cross-domain HSI classification, few-shot learning (FSL) has emerged as a solution. FSL aims to enable rapid learning and generalization of models under conditions of limited labeled data [31]. Liu et al. [32] proposed a deep FSL (DFSL) method to tackle the problem of small sample sizes in HSI classification. By employing three innovative strategies, including the use of a deep residual 3D-CNN to extract spectral–spatial features, training the network in "episodes" to learn a metric space, and classifying test samples in the learned metric space with a nearest neighbor classifier, the method has been proven to achieve better classification accuracy than traditional semi-supervised methods with only a few labeled samples. Gao et al. [33] presented a few-shot hyperspectral classification model based on a relational network, combining the concept of meta-learning to accurately classify new HSIs with a limited number of labeled samples, and demonstrated its superior classification performance over traditional semi-supervised support vector machines (SVMs) and semi-supervised deep learning models on three public datasets. However, few-shot HSI classification methods also have some drawbacks. For some complex HSI data, these methods may require more labeled data to achieve good classification performance. Second, due to the scarcity of labeled data, models are susceptible to overfitting or underfitting, leading to insufficient generalization capability [34].

To address these issues, cross-domain few-shot HSI classification methods integrate the concepts of cross-domain learning and FSL, aiming to tackle the dual challenges of scarce labeled data and domain shift. Li et al. [35] proposed a novel deep cross-domain FSL (DCFSL) approach, which for the first time, addresses FSL and domain adaptation issues within a unified framework. This method utilizes a conditional adversarial domain adaptation strategy to overcome domain shift and simultaneously executes FSL on both source and target classes, capable of discovering transferable knowledge from source classes and learning a discriminative embedding model for target classes. Zhang et al. [36] introduced a graph information aggregation cross-domain FSL (Gia-CFSL) framework, designed to compensate for the shortcomings in existing cross-scene HSI classification, such as the decline in classification performance for new categories and the neglect of nonlocal spatial information, by combining FSL with domain alignment based on graph information aggregation. Ignoring the global information in previous methods, Peng et al. [37] proposed a convolutional transformer-based FSL (CTFSL) method, which for the first time combined CNNs with vision transformers to capture both local and global features through a feature extractor and reduced domain shift through a domain discriminator. Wang et al. [38] proposed a dual-branch domain adaptation FSL method based on domain separation and domain fusion to achieve domain data distribution alignment while maintaining the independence of domain class distributions. Compared with existing cross-domain methods, an additional domain separation learning branch was introduced, which developed

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a domain gate mechanism network structure to learn discriminative knowledge. Cross-domain few-shot HSI classification methods leverage the data discrepancies between the source and target domains, along with the technical means of FSL, enabling accurate classification of HSIs with a limited amount of labeled data and holding significant potential for practical applications. Despite the progress made, existing cross-domain few-shot methods often have certain limitations. First, the single top-level discriminative feature adoption of conditional domain adversarial training strategies to address the domain shift issue between the source and target domains overlooks more complex feature alignment needs, leading to insufficient generalization capabilities. Second, the sample features between domains inherently possess some discriminability, and direct alignment can damage the category information of the target domain features, leading to negative transfer.

To overcome these limitations, this article introduces a new multilevel prototype alignment (MLPA) framework for cross-domain few-shot HSI classification. The MLPA aligns the feature distributions of the source and target domains layer by layer through a multilevel alignment strategy, effectively reducing domain differences and enhancing the model's generalization ability. In addition, the MLPA method designs a prototype alignment module that combines class prototype-based domain adversarial training for the feature alignment process, ensuring that while reducing domain shift, the discriminability of category features is maintained, avoiding the confusion of features between categories.

The main contributions of this article are as follows.

1) A framework for cross-domain few-shot HSI classification called MLPA has been proposed, aiming to effectively align the significant distribution differences between two domains and extract invariant spectral–spatial features. By transferring knowledge from the source domain with sufficient labeled samples to the target domain with limited labeled samples, MLPA can achieve accurate cross-domain classification while maintaining high generalization and discrimination even when the target domain has scarce annotated data.
2) To address the issue of coarse feature representation that may arise from traditional domain alignment methods, an innovative layer-by-layer feature alignment strategy has been introduced. This strategy embeds prototype alignment modules at multiple hierarchical levels within a 3-D residual feature extractor, achieving fine-grained cross-domain alignment of feature distributions from the source and target domains. This layer-by-layer alignment can more comprehensively adjust the feature representations at various layers of the network, thereby enhancing the model's generalization capability on the target domain.
3) To resolve the potential negative transfer issue in the target domain that may result from directly aligning the feature distributions of the source and target domains, a prototype-based domain alignment module has been designed for domain adversarial training. By aligning the central features of each category—namely, the prototypes—while preserving the discriminative nature of the intra-domain class features, it effectively prevents the blurring of class boundaries, enhancing both the discriminative and generalization capabilities of the features in the target domain.

II. Related Works

A. Spectral–Spatial Residual Network

The SSRN enhances classification accuracy by employing a series of consecutive spectral and spatial residual blocks, which are specifically designed to learn the spectral and spatial features of the image. In addition, the SSRN processes spectral and spatial features independently to extract more discriminative characteristics. Finally, the implementation of batch normalization in each convolutional layer significantly accelerates the training convergence of the model.

B. Deep Few-Shot Learning

DFSL aims to address the few-shot problem in HSI classification. The algorithm is implemented through three main strategies. First, a deep residual 3D-CNN is utilized to extract spectral–spatial features, reducing labeling uncertainty. Second, a metric space is learned through episode-based training, ensuring that samples of the same class are close in the space, while those of different classes are distant. Finally, a nearest neighbor classifier is applied within this metric space to classify test samples.

C. Relation Network for Few-Shot Classification

The relation network for few-shot classification (RN-FSC) algorithm is a meta-learning-based few-shot classification model. The algorithm designs a deep classification model that includes modules for feature learning and relational learning. It uses a 3-D convolutional network to extract spatial–spectral features and learns relational scores by comparing sample similarities in the feature space. RN-FSC employs a task-driven learning strategy, training the model by generating a large number of tasks on different datasets, thereby learning transferable feature knowledge and the ability to learn how to learn.

D. Deep Cross-Domain FSL

The DCFSL algorithm is designed to address the cross-domain few-shot problem in HSI classification. The algorithm utilizes a deep residual 3D-CNN to extract spatial–spectral features while performing FSL alternately in the source and target domains. It learns transferable knowledge by training on the source domain data, which then assists the classification task in the target domain. Furthermore, it employs a conditional adversarial domain adaptation strategy to overcome the domain shift between the source and target domains, achieving domain distribution alignment.

E. Graph Information Aggregation Cross-Domain FSL

The Gia-CFSL method is used to address cross-scenario issues in HSI classification, especially when there are sensor and category differences between the source and target

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domains. The algorithm combines FSL with graph information aggregation, representing and aggregating nonlocal spatial relationships through an internal domain distribution extraction block (IDE-block) and a cross-domain similarity-aware block (CSA-block). It performs feature-level and distribution-level cross-domain graph alignment to reduce the impact of domain shift. It utilizes graph optimal transport (GOT) technology to achieve domain distribution alignment and enhances the information exchange between the source and target domains through an attention mechanism.

III. Methodology

In the setting of cross-domain few-shot problems, there are two domain datasets involved: the source domain dataset D and the target domain dataset D_t. The source domain dataset includes C classes, and the target domain dataset also contains C classes. The data categories of the source and target domains may be the same or different, but there are differences in their data distributions. To learn rich feature representations and knowledge from the source domain, the quantity of samples in the source domain is typically larger than that in the target domain, and C is greater than C. For the target domain dataset, it is divided into a few labeled dataset D_f and an unlabeled test dataset D_t, with $D_f \cup D_t = D_T$. D_f and D_t have the same C_t classes, and $D_f \cap D_t = \emptyset$.

A. Overall Architecture

The proposed MLPA framework, as shown in Fig. 1, consists of two mapping layers, a 3-D residual feature extractor, two FSL modules, and three prototype alignment modules. Due to the scarcity of labeled samples in the target domain, which makes it difficult to construct sufficient meta-tasks, the few sample dataset D_f is augmented with random Gaussian noise to increase the sample size. The source and target domains then alternately perform episode training tasks. Specifically, the source and target domains are aligned in input dimension through the source domain's mapping layer and the target domain's mapping layer, respectively. Subsequently, a shared-parameter 3-D residual feature extractor is used to extract the embedded feature vector representation of the samples. FSL is performed in the spectral–spatial embedding space using metric learning methods. Finally, an MLPA module is utilized to progressively achieve domain distribution alignment between the source and target domains, mitigating the effects of domain shift.

This method utilizes a source domain dataset D_S with a large number of labeled samples and a target domain dataset D_f with a small number of labeled samples to train a feature extractor. It then uses the trained feature extractor along with a K-nearest neighbors (KNN) classifier to classify and test the target domain test dataset D_t.

B. Mapping Layer and 3-D Residual Feature Extractor

Each hyperspectral cube of the sample, after passing through the mapping layer and the 3-D residual feature extractor, obtains its spatial–spectral embedded features. The network parameter details of the mapping layer and the 3-D residual feature extractor are shown in Table I. The mapping layer for both the source and target domains consists of a 2-D convolutional kernel of size 1×1, with an input channel number of ch and an output channel number of 100, mapping the source or target domain HSI cube of size $9 \times 9 \times$ ch to data with the same spectral dimension.

TABLE I
Network Parameters of Mapping Layer and Feature Extractor

Layer Name	Input Size	Filter Size	Output Size	Padding
Input	9×9×ch	N/A	N/A	N
Mapping	9×9×ch	1×1×100	9×9×100	N
Convolution1	9×9×100×1	3×3×3×8	9×9×100×8	Y
Convolution2	9×9×100×8	3×3×3×8	9×9×100×8	Y
Convolution3	9×9×100×8	3×3×3×8	9×9×100×8	Y
Shortcut	Convolution1+ Convolution3			
Max Pooling1	9×9×100×8	2×2×4	5×5×25×8	N
Convolution4	5×5×25×8	3×3×3×16	5×5×25×16	Y
Convolution5	5×5×25×16	3×3×3×16	5×5×25×16	Y
Convolution6	5×5×25×16	3×3×3×16	5×5×25×16	Y
Shortcut	Convolution4+ Convolution6			
Max Pooling2	5×5×25×16	2×2×4	3×3×7×16	N
Convolution7	3×3×7×16	3×3×3×32	1×1×5×32	N
Flatten (Output)	1×1×5×32	N/A	160	N

For a fair comparison, the network structure of the 3-D residual feature extractor is consistent with [32] and [35], including two residual blocks, two max-pooling layers, and a 3-D convolutional kernel.

C. MLPA Module

To alleviate the domain shift between the source and target domains, previous methods employed domain adversarial training with a domain discriminator at the last layer of the feature extractor to obtain domain-invariant features [35], thereby achieving domain alignment. However, this approach has some limitations; domain alignment at only the last layer may struggle to capture and adjust the feature representations of all relevant layers throughout the network, limiting the model's generalization ability on the target domain. In the feature extraction network, shallow features often contain more universal information, while higher level features may contain more task-specific information as the network depth increases, leading to performance degradation when the model is transferred between the source and target domains. To address this issue, inspired by [39], the MLPA framework adopts a layer-by-layer alignment strategy using a prototype alignment module at different levels of the 3-D residual feature extractor to align the feature distributions between the source and target domains layer by layer, as shown in Fig. 2. By cross-domain aligning the shallow, middle, and high-level features layer by layer, the network can more finely adjust the feature representations, resulting in final feature representations that are more generalized on the target domain.

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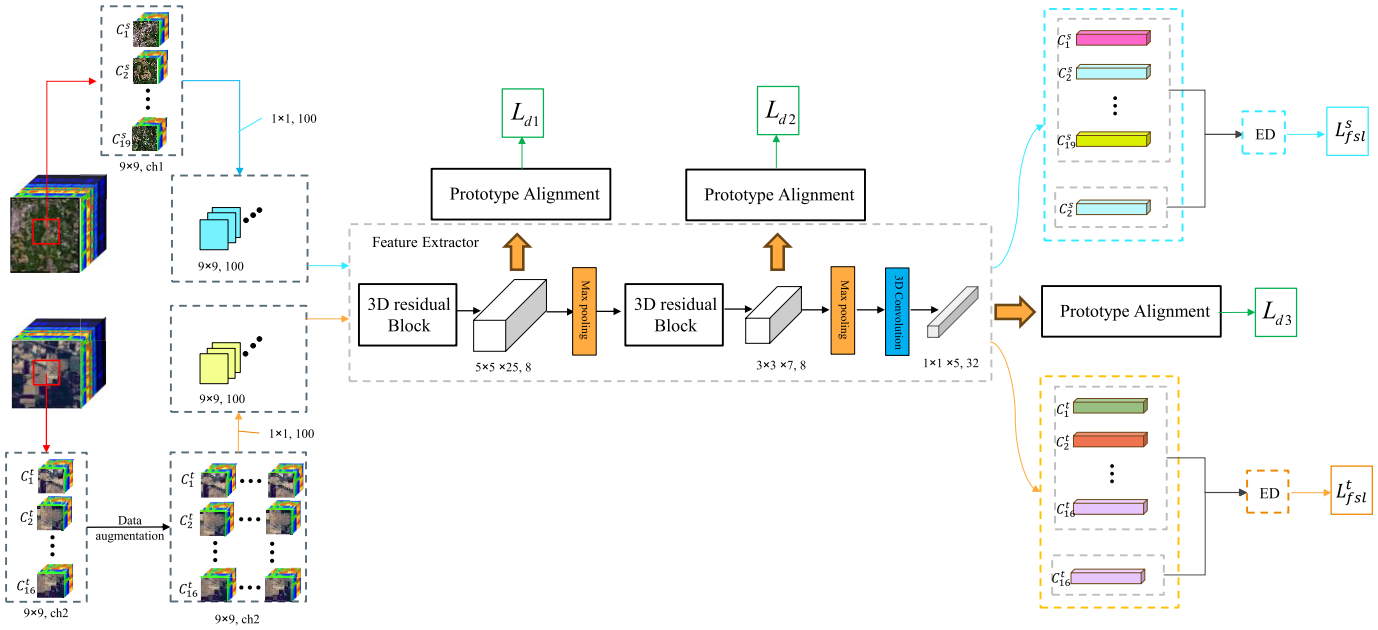


Fig. 1. Framework of the proposed MLPA in the training phase.

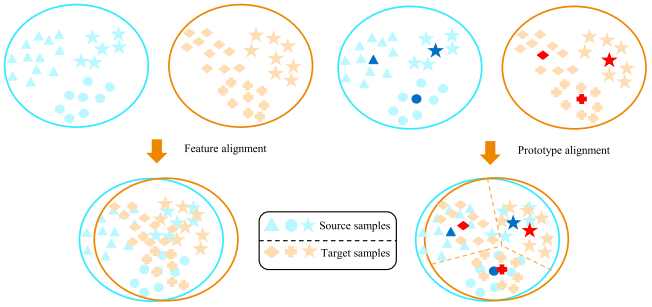


Fig. 2. Cross-domain prototype feature alignment diagram.

While aligning the feature distributions of the source and target domains, directly aligning the sample features of the source and target domains can, to some extent, cause the originally well-distinguished category feature distributions within the domain to overlap, leading to a blurring of the boundaries between categories, as shown in the left of Fig. 2. To improve the generalizability of the target domain features while ensuring their discriminability, a prototype-based domain alignment module was designed for domain adversarial training. As shown in the right of Fig. 2, class prototypes represent the central features of each category. Aligning the class prototype features of the source and target domains helps maintain the feature distributions of the categories within the domain without significant changes, preserving the unique discriminative information on the target domain. The structure of the prototype alignment module is shown in Fig. 3. Taking the shallow-layer prototype alignment as an example, the shallow features of the C-way K-shot samples from the source and target domains first undergo a global average pooling to obtain the corresponding feature vector representations. Then, the average of the feature vectors for the same category in the source and target domains is taken to obtain the prototype vector representing this category. Finally, the prototype feature vectors of the source and target domains are input into the domain discriminator D, which consists of five fully connected layers, to distinguish the feature representations of the source and target domains. The domain adversarial loss is defined on D, with the formula being

$$L_{d_n} = \min_{D_n} \max_{F_n} L = -E_{z\sim C_s} \log[D_n(\mu_z^s)] - E_{c\sim C_t} \log[1 - D_n(\mu_c^t)] \quad (1)$$

where μ_z^s and μ_c^t represent the class prototype features of the source and target domains obtained by the feature extraction network F_n at the n-th layer, respectively. $D_n(\cdot)$ denotes the probability predicted by the discriminator at the n-th layer as the source domain, while $1 - D_n(\cdot)$ represents the probability predicted as the target domain. While the feature extractor F aims to generate domain-invariant feature representations by maximizing the domain adversarial loss L, D seeks to minimize the domain adversarial loss to distinguish between the feature representations, making the features generated by F indistinguishable between the source and target domains, thus achieving domain alignment.

D. Training Paradigm

During the training process, a cross-domain alternating fine-tuning training paradigm is used to optimize the network parameters, that is, in each iteration, full-parameter episodic training is first conducted on the source domain data, and then, during episodic training on the target domain, the early layers of the feature extractor are frozen, while only the last layer is fine-tuned, as shown in Fig. 4. The feature distributions of the source and target domains are aligned layer by layer through a domain adversarial training strategy. The entire training phase is shown in Algorithm 1. By conducting alternating fine-tuning episodic training on the source and target domain data, the

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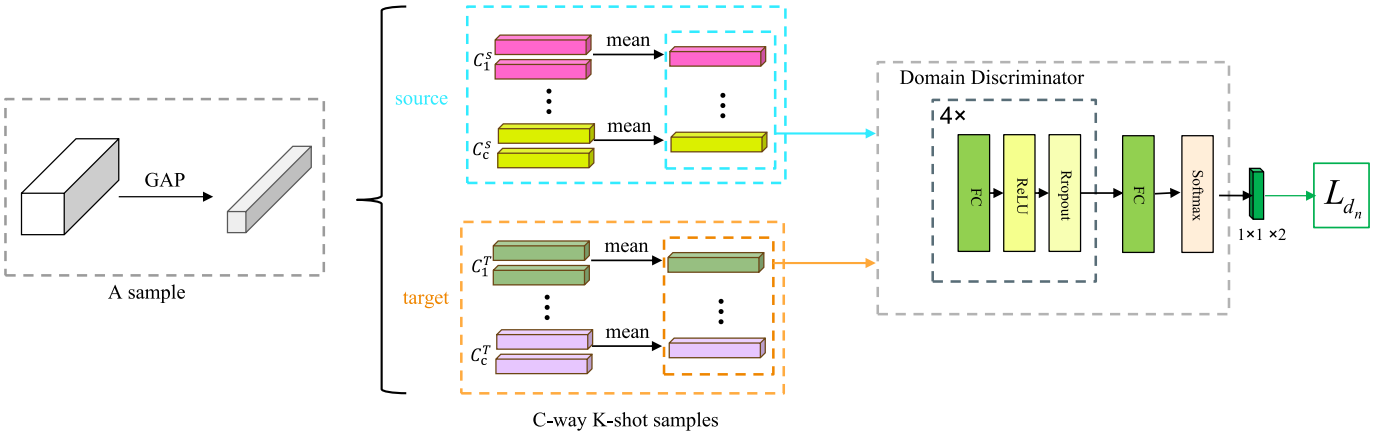


Fig. 3. Structure of prototype alignment module.

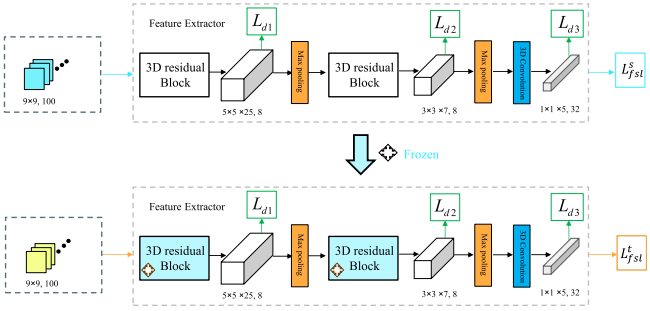


Fig. 4. Flowchart of cross-domain alternating fine-tuning training.

features extracted by the feature extractor are endowed with both strong generalization and discrimination.

Taking the source domain FSL as an example, a training task for an episode randomly selects C classes and, for each class, selects $K + N$ $9 \times 9 \times$ ch1 HSI cubes as model input, where the labeled support set $S_s = \{(x_i, y_i)\}_{i=1}^{C\times K}$ and the unlabeled query set $Q_s = \{(x_j, y_j)\}_{j=1}^{C\times N}$ are included. By passing the samples through the mapping layer and feature extractor F, they are mapped into the feature space. Then, a distance-based softmax activation function is used to calculate the predicted probabilities for each sample in the query set

$$P(y_j = k|x_j \in Q_s) = \frac{\exp(-d(z_j, z_{\text{mean}(k)}))}{\sum_k^C \exp(-d(z_j, z_{\text{mean}(k)}))} \tag{2}$$

where $d(,)$ represents the Euclidean distance function, x_j is the jth sample in the query set, y_j is its predicted label, C is the number of classes in an episode task, z_j is the embedded feature of the query set, and $z_{\text{mean}(k)}$ is the average embedded feature of the support set for the kth class. Therefore, the FSL loss for the source domain can be formulated as follows:

$$L_{\text{fsl}}^s = E_{S_s,Q_s}\left[-\sum_{(x,y)\in Q_s} \log p_\varphi(y = k|x)\right] \tag{3}$$

where S_s and Q_s represent the support set and query set in the source domain dataset, respectively. Similarly, the FSL loss L_{fsl}^t for the target domain is consistent with that of the source domain FSL.

Algorithm 1 Training Phase of MLPA

Input:
The source dataset D_s
The target dataset D_t
The support set S_s and query set Q_s from D_s
The support set S_t and query set Q_t from D_t
Number of training episodes N

Output: The total loss L_{total} of the model

1: **for** episode = 0: N-1 **do**
2: **if** episode%2 = = 0 **then**
3: **for** each param in feature_encoder.parameters() **do**
4: Unfreeze
5: **end for**
6: Train with S_s, Q_s and D_t
7: Calculate prototype features of the n-th layer's $S_s \rightarrow Z_s^n$ for FSL (n = 3) and Domain Adaptation (n = 1,2,3)
8: Calculate prototype features of the n-th layer's $D_t \rightarrow \mu_t^n$ for Domain Adaptation (n = 1,2,3)
9: Calculate L_{fsl}^s of an episode by Ep. (3)
10: Calculate L_{d_n} and L_d of an episode by Ep. (1) and Ep. (4)
11: Calculate L_{total}^s of an episode by Ep. (5)
12: L_{total}^s.backward( )
13: **else**
14: **for** each param in block.parameters() **do**
15: Freeze
16: **end for**
17: Train with the same process as above as using S_t, Q_t and D_s
18: **end for**

The loss generated by the MLPA module through layer-by-layer domain adversarial training is expressed as

$$L_d = L_{d_1} + L_{d_2} + L_{d_3}. \tag{4}$$

Therefore, the total loss for the source domain is expressed as

$$L_{\text{total}}^s = L_{\text{fsl}}^s + L_d. \tag{5}$$

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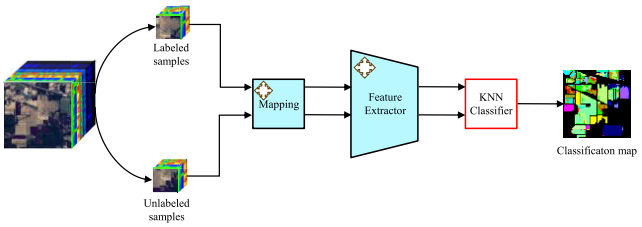


Fig. 5. Flowchart of classification in the testing phase.

The total loss for the target domain is expressed as

$$L^t_{\text{total}} = L^t_{\text{fsl}} + L_d. \tag{6}$$

Ultimately, the loss function for MLPA is formulated as

$$L_{\text{total}} = L^s_{\text{total}} + L^t_{\text{total}}. \tag{7}$$

E. Testing Paradigm

During the testing stage, as the trained mapping layer and feature extractor have learned the ability to model the distances between samples of different categories in the metric space, namely, samples of the same category are close and those of different categories are distant, they can serve as embedding functions to map samples into the Euclidean metric space. As KNN is a nonparametric classification algorithm, it can directly utilize this mapped feature space for classification decisions. The process of classifying unlabeled samples in the target domain using the trained mapping layer, feature extractor, and KNN classifier (NN) is depicted in Fig. 5. First, the labeled few-shot dataset from the target domain and the remaining unlabeled samples pass through the mapping layer and feature extractor to obtain feature vectors with high interclass discriminability. Then, the NN is fit using the feature vectors of the labeled samples. Finally, the NN is used to classify the unlabeled samples.

IV. Experiments

A. Data Description

To evaluate the effectiveness of the MLPA method, the experiments utilize four public datasets, namely, Chikusei (CH), Indian Pines (IP), the University of Pavia (UP), and Houston (HT). The Chikusei dataset is used as the source domain, while the other datasets serve as target domains. The specific details of the experimental datasets are shown in Table II.

Chikusei [40]: The dataset was collected on July 29, 2014, in Chikusei City, Japan, by a hyperspectral visible/near-infrared camera (hyperspecv-vnir-c). It contains 19 classes with a size of 2517 × 2335 pixels. Its spatial resolution is 2.5 m/pixel. It consists of 128 spectral bands ranging from 363 to 1018 nm. A false-color image of the Chikusei dataset and the corresponding ground-truth map are shown in Fig. 6.

Indian Pines (IP): The IP dataset was also collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) in 1992 from northwestern Indiana. It consists of 200 bands, ranging from 400 to 2500 nm. It has a size of 145 × 145 pixels. Its spatial resolution is 20 m/pixel. It includes 16 vegetation classes. A false-color image of the IP dataset and the corresponding ground-truth map are shown in Fig. 7.

The University of Pavia (UP): The UP dataset was collected by the Reflective Optics System Imaging Spectrometer (ROSIS). It contains 610 × 340 pixels. Its spatial resolution is 1.3 m/pixel. It has 103 spectral bands, ranging from 430 to 860 nm. There are nine classes representing different land cover types. A false-color image of the UP dataset and the corresponding ground-truth map are shown in Fig. 8.

Houston (HT): The HT dataset was acquired by the ITRES CASI-1500 sensor and provided by the 2013 IEEE GRSS Data Fusion Contest. The data size is 349 × 1905, containing 144 spectral bands ranging from 364 to 1046 nm. It includes 15 classes of land cover types. A false-color image of the HT dataset and the corresponding ground-truth map are shown in Fig. 9.

B. Experimental Setup

On the Ubuntu 20.04 operating system, experiments were conducted using an RTX3090 graphics card with 24 GB of video memory. The deep learning framework selected was PyTorch 1.12.1, with CUDA version 11.3, Python version 3.8, and the Adam optimizer was used. Following the experimental setup of [35], the training setup had an initial learning rate of 0.001, and the model was trained for a total of 10 000 iterations. The spatial size of the input cube was selected to be a 9 × 9 neighborhood, and all convolutional layer parameters were initialized using the Xavier normalization. During the meta-training phase, the number of categories for the support set and query set was set to the number of categories in the target domain dataset (e.g., nine for the UP set and 16 for the IP set), with one sample per category for the support set and 19 for the query set. Theoretically, the more samples per category in the query set, the stronger the model's generalization ability, and the more stable the training results [41]. In the testing phase, a NN was fit for each category in the target domain using five samples to test the test set. To avoid the impact of random sampling, the same fixed random seed as [35] and [36] was used, and each experiment was performed ten times to calculate the mean and standard deviation of the overall accuracy (OA), average accuracy (AA), and Kappa coefficient (kappa).

C. Quantitative Comparisons

To verify the effectiveness of the MLPA method for HSI classification, comparative experiments were conducted with methods such as SVM [42], 3DCNN, SSRN, DFSL + NN, RN-FSC, DCFSL, and Gia-CFSL. The experimental results of the five earlier hyperspectral classification methods, namely, SVM, 3DCNN, SSRN, DFSL, and RN-FSC, were referenced from the results reported in [35]. Tables III and IV show the five-shot comparison results of these eight methods on the IP and UP datasets, respectively. Since DFSL + NN and RN-FSC were trained on the HT dataset in the source domain, they could not be tested as results for the target domain dataset with limited samples. Table V shows the five-shot comparison results of the remaining six methods on the HT dataset.

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TABLE II
ALL DETAILS OF THE EXPERIMENTAL DATASETS

	Chikusei	IP	UP	HT
Spatial size	2517×2335	145×145	610×340	349 x 1905
No. of bands	128	200	103	144
Spectral range	363-1018	400-2500	430-860	364-1046
Sensor type	Hyperspec-VNIR-C	AWIRIS	ROSIS	ITRESC ASI-1500
Total classes	19	16	9	15
Labeled samples	77592	10249	42776	15029

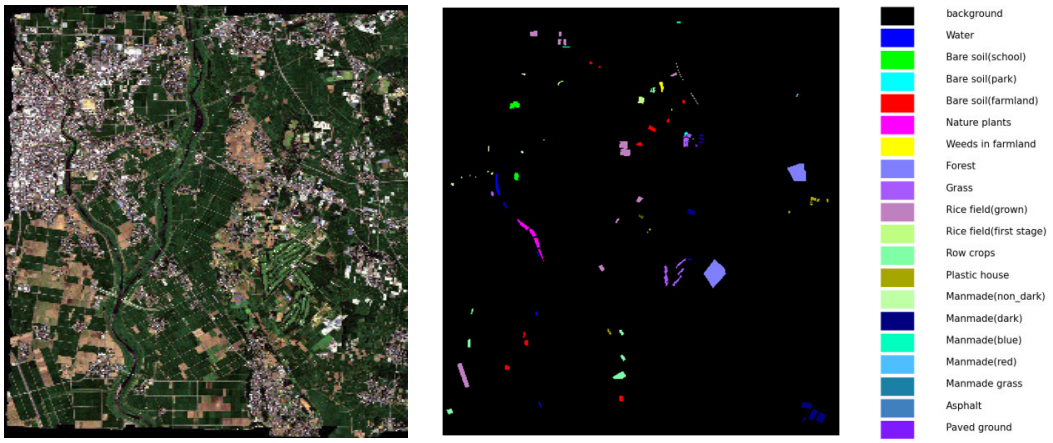


Fig. 6. Chikusei dataset. (a) Pseudo-color image, (b) ground-truth map, and (c) category of CH.

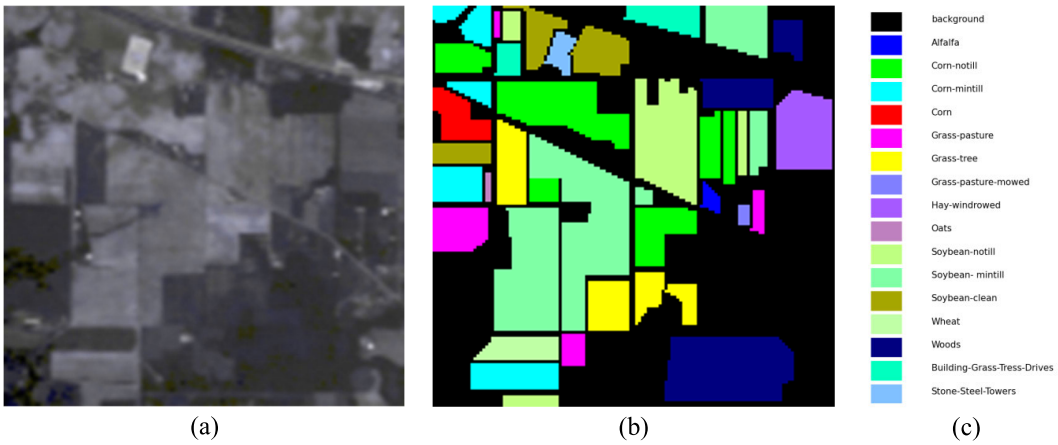


Fig. 7. India Pines dataset. (a) Pseudo-color image, (b) ground-truth map, and (c) category of IP.

It can be observed that, on the IP dataset, the MLPA method outperforms DCFSL, another domain adaptation-based cross-domain few-shot method, by 5.82, 4.65, and 6.54 in OA, AA, and Kappa metrics, respectively; and outperforms Gia-CFSL by 6.73, 5.96, and 7.74 in OA, AA, and Kappa, respectively. On the UP dataset, the MLPA method outperforms DCFSL by 1.82, 0.45, and 2.11 in OA, AA, and Kappa, respectively; and outperforms Gia-CFSL by 0.99, 0.44, and 1.24 in OA, AA, and Kappa, respectively. On the HT dataset, the MLPA method outperforms DCFSL by 1.99, 1.9, and 2.16 in OA, AA, and Kappa, respectively; and outperforms Gia-CFSL by 3.95, 3.89, and 4.27 in OA, AA, and Kappa, respectively.

Among these methods, SVM, as a traditional classification method, focuses only on spectral features and neglects spatial information. Moreover, when facing the challenges of high-dimensional spectral features and few labeled samples, its classification accuracy is the lowest. Compared to traditional methods, 3DCNN and SSRN, which are based on deep convolutional networks, can comprehensively extract spatial and spectral features from images, effectively capturing complex patterns in hyperspectral data. However, in the case of few samples, the models are prone to overfitting, are difficult to train adequately, and thus perform poorly. Facing the situation of few labeled samples, DFSL + NN and RN-FSC have

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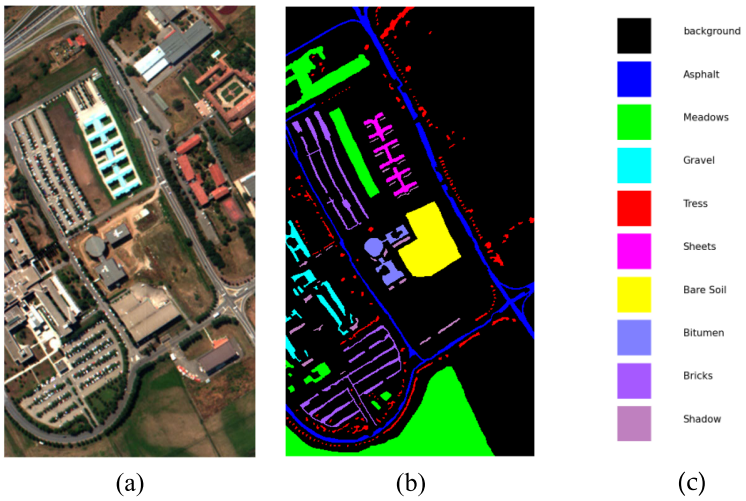


Fig. 8. UP dataset. (a) Pseudo-color image, (b) ground-truth map, and (c) category of UP.

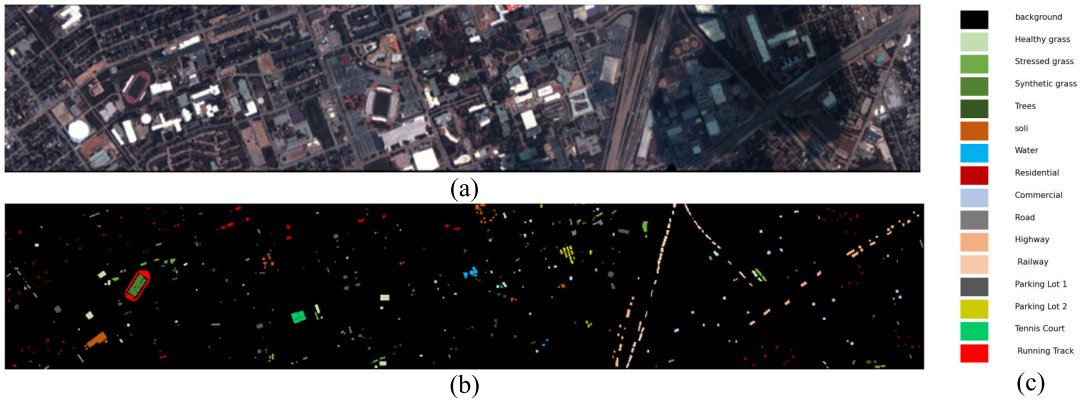


Fig. 9. HT dataset. (a) Pseudo-color image, (b) ground-truth map, and (c) category of HT.

good classification accuracy. This is because they perform meta-learning on the source domain to capture a broader range of feature representations, which also have good generalization capabilities on the target domain, enhancing their performance there. DCFSL, as a domain adaptation-based cross-domain few-shot method, uses conditional adversarial training strategies to align the feature distributions of the source and target domains, enhancing the generalizability of the target domain features. The Gia-FSL method considers nonlocal spatial information with strong correspondence when performing domain alignment and uses feature-level and distribution-level cross-domain graph alignment to mitigate the impact of domain shift on FSL, thus achieving better classification results. The MLPA method, by using an MLPA module, aligns the feature distribution differences between the source and target domains at a finer granularity and mitigates the negative transfer problem, further enhancing the performance of cross-domain few-shot methods with domain adaptation. It has achieved the best classification results among these methods.

D. Qualitative Comparisons

Figs. 10–12 visualize the classification maps of the three domain adaptation few-shot methods, DCFSL, Gia-CFSL, and MLPA, on the IP, UP, and HT datasets. The results generated by the proposed MLPA are the closest to the ground-truth maps, especially on the IP dataset. The IP dataset poses significant challenges due to its high number of categories, imbalanced sample distribution, and high noise levels, making classification difficult. However, MLPA enhances the model's generalization and discrimination capabilities, resulting in stronger robustness and the most noticeable classification effects on the IP dataset. The UP dataset has relatively fewer categories, and the few-shot domain adaptation methods all show significant classification effects on the UP dataset. Compared to the other two methods, MLPA has achieved some improvement in the classification results for the second category "Meadows" and the third category "Gravel." In the HT dataset, the labeled categories are relatively dispersed, and MLPA is closer to the true label map for the tenth category "Highway." In summary, the visualized classification results further demonstrate the superiority of the MLPA method.

In addition, to more intuitively compare the feature extraction effects of DCFSL, Gia-CFSL, and MLPA, these three domain adaptation methods on the IP, UP, and HT datasets, the t-SNE technique is used to map high-dimensional features of different category samples into a 2-D space for visualization, as shown in Fig. 13. It is evident that the proposed MLPA

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TABLE III
CLASSIFICATION RESULTS (%) ON THE IP DATASET (FIVE SHOTS)

Class	SVM	3DCNN	SSRN	DFSL+ NN	RN-FSC	DCFSL	Gia-CFSL	Ours
1	72.20	95.12	18.38	96.75	96.34	95.85	94.63	**98.54**
2	34.27	37.70	**64.79**	38.65	46.13	41.88	40.43	57.08
3	39.18	19.77	27.65	42.79	40.61	47.96	47.35	**60.27**
4	50.34	32.51	26.97	68.10	58.62	79.53	77.89	**83.15**
5	69.75	**88.45**	80.76	71.20	64.96	71.76	74.64	75.96
6	66.36	73.65	86.87	76.18	69.45	84.72	82.51	**86.95**
7	89.13	81.82	32.24	**100.00**	**100.00**	99.13	99.57	99.13
8	68.73	53.35	**100**	74.84	77.70	83.70	81.48	90.70
9	86.67	100	57.69	**100.00**	**100.00**	99.33	99.33	98.67
10	37.49	41.35	59.69	47.98	25.49	60.94	58.04	**68.49**
11	33.96	66.71	**70.87**	57.95	65.51	59.47	62.56	59.84
12	31.43	37.40	45.00	38.21	27.13	46.68	44.49	**55.58**
13	86.50	85.71	88.29	97.50	**99.75**	97.75	96.85	97.20
14	62.93	62.57	**97.18**	83.44	76.35	84.71	80.87	86.63
15	28.08	56.42	36.64	62.29	70.34	68.45	59.40	**78.45**
16	90.91	90.36	60.98	100	100	98.64	**99.43**	98.30
OA	45.85 ± 2.44	54.76 ± 0.03	61.36 ± 0.49	59.65 ± 0.63	58.17 ± 0.02	64.29 ± 2.76	63.38 ± 2.71	**70.11 ± 2.62**
AA	59.24 ± 1.36	63.93 ± 0.02	59.75 ± 0.20	72.24 ± 0.42	69.90 ± 0.40	76.28 ± 1.24	74.97 ± 2.05	**80.93 ± 1.58**
Kappa	39.68 ± 2.48	48.72 ± 0.03	56.91 ± 0.48	54.55 ± 0.52	52.52 ± 0.14	59.87 ± 2.82	58.67 ± 2.88	**66.41 ± 2.76**

TABLE IV
CLASSIFICATION RESULTS (%) ON THE UP DATASET (FIVE SHOTS)

Class	SVM	3DCNN	SSRN	DFSL+ NN	RN-FSC	DCFSL	Gia-CFSL	Ours
1	88.98	59.82	**91.84**	69.19	68.55	78.09	73.25	80.33
2	83.91	63.05	**95.13**	84.63	93.44	81.63	84.95	85.50
3	39.98	68.91	55.23	57.47	49.81	64.72	62.70	**65.01**
4	60.22	77.31	78.02	89.99	92.15	**94.65**	94.05	93.24
5	95.44	90.77	98.34	**100**	99.43	99.38	99.65	99.07
6	37.12	63.40	53.56	71.23	57.99	**74.94**	73.73	73.74
7	40.62	87.64	60.07	70.62	70.04	78.44	**81.94**	78.09
8	68.17	57.27	85.34	58.13	63.48	68.25	**72.34**	68.62
9	99.13	95.57	98.08	96.92	99.19	98.62	96.18	**99.25**
OA	64.12 ± 4.55	65.74 ± 1.77	76.26 ± 5.78	77.75 ± 1.16	80.19 ± 2.18	80.08 ± 3.92	80.91 ± 1.29	**81.90 ± 3.45**
AA	68.18 ± 2.27	73.72 ± 1.01	79.51 ± 3.21	77.57 ± 0.31	77.12 ± 0.84	82.08 ± 1.91	82.09 ± 2.04	**82.53 ± 1.72**
Kappa	55.59 ± 4.64	57.37 ± 1.97	70.56 ± 6.69	71.11 ± 1.22	73.73 ± 2.79	74.46 ± 4.45	75.33 ± 1.53	**76.57 ± 4.09**

method achieves a higher degree of feature separation across different categories and more compact clustering within the same category compared to DCFSL and Gia-CFSL on all three datasets, demonstrating the optimal classification boundaries. For instance, on the IP dataset, categories 2, 3, 11, and 12 are intermingled and difficult to distinguish. In Fig. 13(a)–(c), it can be seen that the proposed method provides more compact clustering for these four categories and clearer decision

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TABLE V
CLASSIFICATION RESULTS (%) ON THE HT DATASET (FIVE SHOTS)

Class	SVM	3DCNN	SSRN	DCFSL	Gia-CFSL	Ours
1	81.03	89.24	81.94	90.20	89.66	**90.70**
2	74.54	77.15	**80.72**	74.45	74.43	80.46
3	97.60	95.06	**99.68**	94.65	93.48	95.34
4	86.28	88.15	89.81	91.38	89.18	**91.83**
5	89.06	94.68	95.78	**96.90**	94.69	95.98
6	82.52	72.12	83.57	84.34	79.61	**86.14**
7	55.96	54.53	66.79	59.29	58.38	**67.87**
8	24.85	42.38	22.36	42.59	37.74	**44.54**
9	58.49	59.98	**61.50**	59.95	55.93	58.10
10	47.67	53.29	49.14	66.79	60.37	**73.54**
11	51.04	61.13	**77.97**	56.02	59.46	57.08
12	39.67	44.77	40.70	**75.42**	71.84	73.40
13	19.61	31.56	75.09	86.73	88.64	**92.57**
14	93.48	86.36	**99.80**	88.60	87.42	88.27
15	96.05	96.56	**99.77**	94.65	91.41	94.79
OA	63.75 ± 2.26	68.58 ± 2.58	70.46 ± 1.24	74.22 ± 2.92	72.26 ± 3.60	**76.21 ± 2.46**
AA	66.52 ± 1.91	69.80 ± 2.61	74.97 ± 0.99	77.47 ± 2.62	75.48 ± 2.82	**79.37 ± 2.25**
Kappa	60.89 ± 2.39	65.92 ± 2.76	68.08 ± 1.33	72.16 ± 3.16	70.05 ± 3.89	**74.32 ± 2.66**

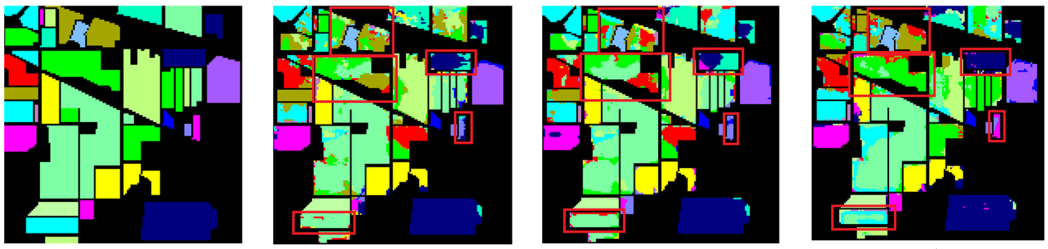


Fig. 10. Visualization results of various methods on the IP dataset. (a) Ground truth, (b) DCFSL, (c) Gia-CFSL, and (d) MLPA.

boundaries. On the UP dataset, categories 3 and 8 are almost overlapping. In Fig. 13(d)–(f), it can be observed that the proposed method alleviates the overlapping situation between these two categories. On the HT dataset, categories 7, 8, 9, and 11 are highly confusable. In Fig. 13(g)–(i), it is evident that the proposed method offers better feature separability among these four categories. In conclusion, the proposed MLPA method effectively enhances the discriminability of features across different categories and the clustering quality of features within the same category.

E. Ablation Experiments

1) Analysis of MLPA Effects: The main contribution of the proposed MLPA lies in the MLPA module, which includes multilevel alignment strategies and prototype alignment modules. To better analyze the role of this module, its function was evaluated through ablation studies, focusing on four scenarios: 1) domain alignment using a domain discriminator with the same structure only at the top layer of the feature extractor (DA); 2) a single prototype alignment module: domain alignment using one prototype alignment module at the top layer (PA3), middle layer (PA2), or shallow layer (PA1) of the feature extractor; 3) two prototype alignment modules: domain alignment using two prototype alignment modules at the top and middle layers (PA3&PA2), top and shallow layers (PA3&PA1), or middle and shallow layers (PA2&PA1) of the feature extractor; and 4) three prototype alignment modules: domain alignment using three prototype alignment modules at the top, middle, and shallow layers (PA3&PA2&PA1) of the feature extractor.

As shown in Table VI, using a single prototype alignment module achieved the highest OA of 65.00 on IP, 81.27 on UP, and 73.79 on HT; using two prototype alignment modules achieved the highest OA of 66.18 on IP, 81.56 on UP, and 74.03 on HT; and using three prototype alignment modules

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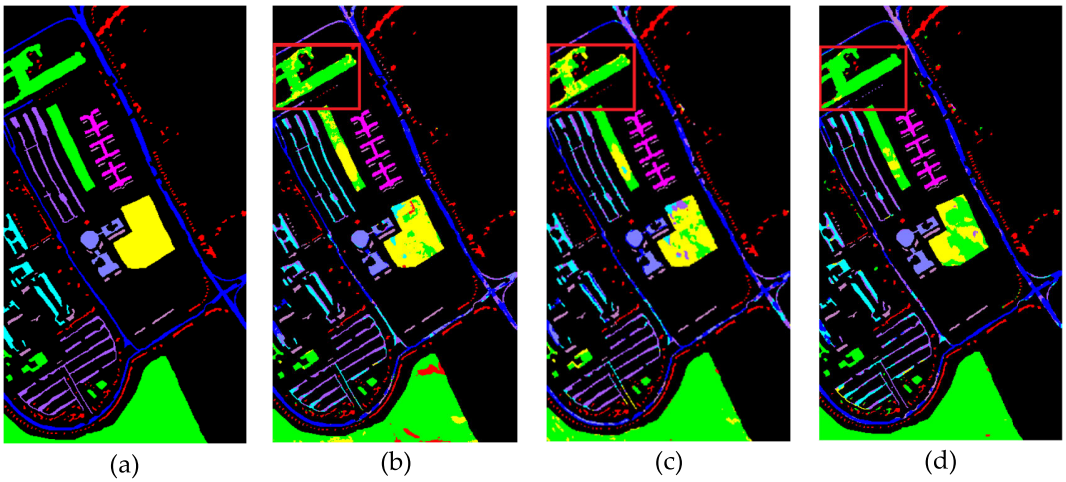


Fig. 11. Visualization results of various methods on the UP dataset. (a) Ground truth, (b) DCFSL, (c) Gia-CFSL, and (d) MLPA.

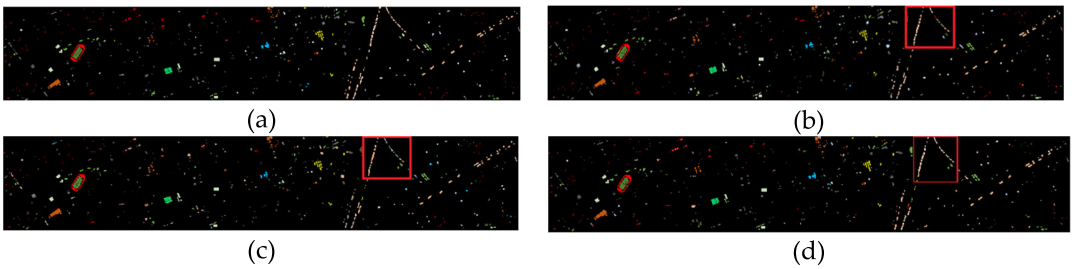


Fig. 12. Visualization results of various methods on the HT dataset. (a) Ground truth, (b) DCFSL, (c) Gia-CFSL, and (d) MLPA.

TABLE VI
COMPARISON OF ABLATION RESULTS (%) OF MLPA

	Module				Dataset		
	DA	PA3	PA2	PA1	IP	UP	HT
1	√				64.45	80.44	73.50
2		√			64.75	81.27	73.38
			√		65.00	80.86	73.69
				√	64.47	80.08	73.79
3		√	√		64.69	81.32	74.03
		√		√	66.13	81.44	73.37
			√	√	66.18	81.56	74.03
4		√	√	√	**67.10**	**81.84**	**75.32**

achieved 67.10 on IP, 81.84 on UP, and 75.32 on HT. It can be seen that: 1) using a prototype alignment module at the top layer compared to a general discriminator for domain alignment increased the OA by 0.3 on IP and 0.83 on UP datasets, proving that prototype alignment can to some extent alleviate negative transfer and 2) using two prototype alignment modules increased the OA by 1.18 on IP, 0.29 on UP, and 0.24 on HT compared to using a single prototype alignment module; and using three prototype alignment modules increased the OA by 0.92 on IP, 0.28 on UP, and 1.29 on HT compared to using two prototype alignment modules. This proves that the multilevel alignment strategy can more effectively align the feature distributions of the source and target domain.

TABLE VII
CLASSIFICATION PERFORMANCE (%) OF DIFFERENT TRAINING STRATEGIES IN MLPA

Training paradigm	IP	UP	HT
Frozen-res1	66.30	81.58	74.40
Frozen-res2	61.90	80.79	74.92
Frozen1	68.19	81.22	74.60
Frozen2	**70.11**	**81.90**	**76.21**

2) Impact of Training Strategies: To explore the impact of different fine-tuning training paradigms on the performance of the MLPA method, different parameter layers were frozen during fine-tuning on the target domain to seek the optimal training paradigm, which mainly includes four parameter freezing strategies: 1) freezing the parameters of the residual connections in the shallow 3-D residual blocks; 2) freezing the parameters of the residual connections in both shallow and middle 3-D residual blocks; 3) freezing the entire shallow 3-D residual blocks; and 4) freezing the 3-D residual blocks in both shallow and middle layers.

As shown in Table VII, fine-tuning paradigm (1) reduced the OA on the IP, UP, and HT datasets by 0.8, 0.26, and 0.92, respectively, compared to full-parameter training; fine-tuning paradigm (2) reduced the OA on the IP, UP, and HT datasets by 5.2, 1.05, and 0.4, respectively, compared to full-parameter training; fine-tuning paradigm (3) increased the OA on the IP dataset by 1.09 and reduced it on the UP and HT datasets

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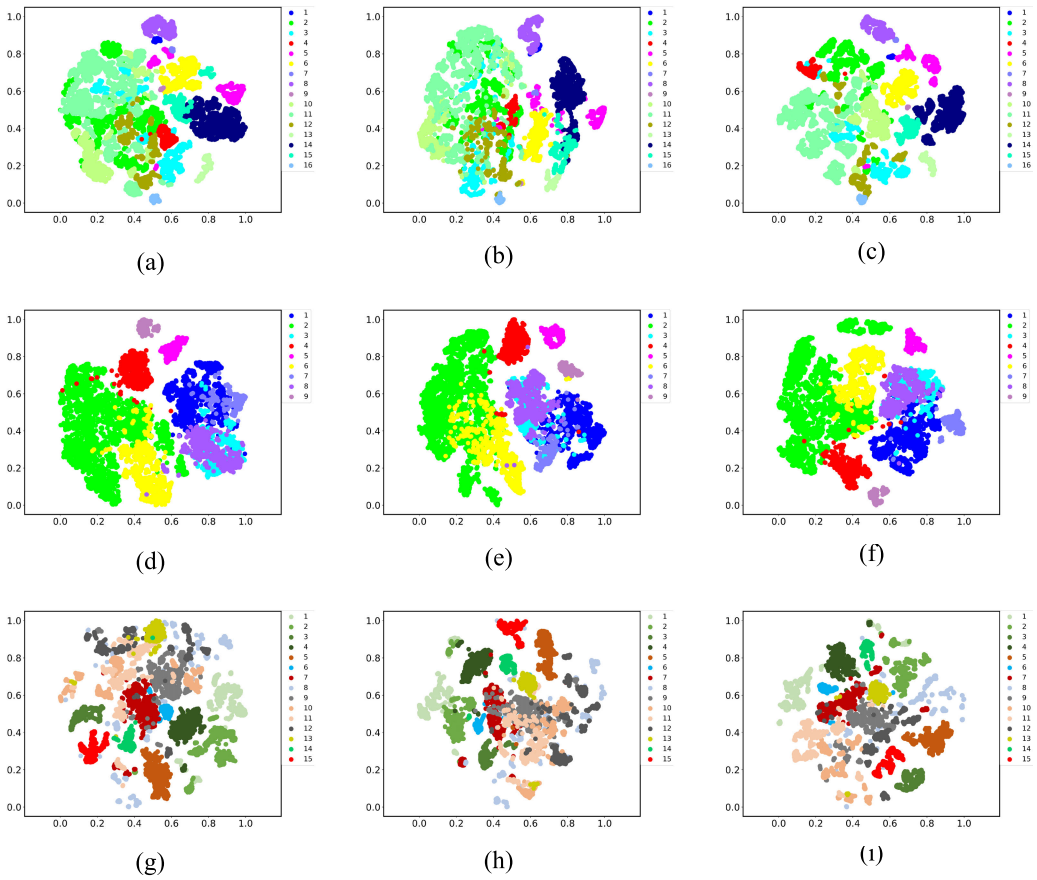


Fig. 13. Two-dimensional feature visualization on three datasets. (a) DCFSL features on IP. (b) Gia-CFSL features on IP. (c) MLPA features on IP. (d) DCFSL features on UP. (e) Gia-CFSL features on UP. (f) MLPA features on UP. (g) DCFSL features on HT. (h) Gia-CFSL features on HT. (i) MLPA features on HT.

TABLE VIII
COMPARE METHODS TO CALCULATE TIME, FLOPS, AND PARAMETERS FOR EACH DATASET

Dataset	Metrics	Method				
		DFSL+NN	RN-FSC	DCFSL	Gia-CFSL	MLPA
IP	Training time	1254.58s	675.21s	2840.01s	9382.18s	2202.40s
	Testing time	1.53s	45.14s	1.93s	1.06s	1.01s
	#Params	0.03M	0.19M	4.26M	0.86M	12.66M
UP	Training time	1032.31s	411.34s	1534.98s	4001.86s	1849.13s
	Testing time	6.74s	47.23s	6.52s	5.33s	7.65s
	#Params	0.03M	0.19M	4.25M	0.86M	12.65M
HT	Training time	-	-	5851.67s	5952.41s	3347.71s
	Testing time	-	-	4.34s	1.51s	2.59s
	#Params	-	-	4.25M	0.86M	12.65M

by 0.62 and 0.72, respectively, compared to full-parameter training; and fine-tuning paradigm (4) increased the OA on the IP, UP, and HT datasets by 3.01, 0.06, and 0.89, respectively, compared to full-parameter training. It can be seen that different fine-tuning strategies have a significant impact on the performance of the MLPA method. Among them, the fine-tuning paradigm (4) significantly improved the performance of the MLPA method, indicating that freezing the residual blocks

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of the first few layers and only fine-tuning the last layer can retain the generalization knowledge from the source domain and effectively prevent overfitting on the few-shot dataset in the target domain. In contrast, the fine-tuning paradigm (2) saw a decline in performance across all datasets, suggesting that excessively freezing the parameters of the residual connections can impair the discriminative ability of the MLPA method in the target domain.

F. Computational Complexity Analysis

To demonstrate the computational efficiency of different few-shot hyperspectral classification methods, Table VIII lists the training time, testing time, and model parameter count for the DFSL + NN, RN-FSC, DCFSL, Gia-CFSL, and MLPA methods on three datasets. It can be observed that the domain adaptation methods DCFSL, Gia-CFSL, and MLPA have a significantly larger number of model parameters and training time compared to DFSL + NN and RN-FSC. The lengthy training time is due to the adversarial domain training process, which requires training a large number of parameters. Among them, the MLPA method, due to the use of multiple prototype alignment modules, has a relatively large number of fully connected layer parameters, resulting in a model parameter count of 12.66 M. However, the training time is significantly lower than the domain adaptation methods DCFSL and Gia-CFSL. This is mainly because the domain alignment process in MLPA trains model parameters using a small number of prototype feature vectors. In contrast, DCFSL and Gia-CFSL train the model using a large number of sample feature vectors, thus greatly reducing the training time.

V. Conclusion

In addressing the challenge of cross-domain few-shot classification in hyperspectral imagery, this article proposes an innovative MLPA method, effectively resolving the distribution discrepancy between the source and target domains while enhancing the model's generalizability and discriminability on the target domain. The core of the MLPA method lies in its layer-by-layer feature alignment strategy and class prototype-based adversarial domain training. Specifically, the layer-by-layer feature alignment strategy achieves fine-grained feature adjustment, ensuring that feature representations from shallow to deep layers can adapt to the data distribution of the target domain, thereby strengthening the model's generalization ability for target domain data. The class prototype-based adversarial domain training reduces intradomain feature distribution overlap and optimizes the discriminability of features in the target domain. Extensive experimental results on the IP, UP, and HT datasets demonstrate that the proposed method effectively mitigates negative transfer and enhances model generalization capability compared to other domain adaptation methods.

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